A-A-002544
April 23, 1987

INTERIM COMMERCIAL ITEM DESCRIPTION
COATING, RUST PREVENTIVE FOR INACTIVE FLEET SHIPS

This interim commercial item description was developed by NAVSEA 070R (C.P.E.C - Norfolk Naval Shipyard, Portsmouth, VA), based on currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for change to the preparing activity at the address shown above.

The coating covered by this purchase description is intended for use on inactive fleet ships and boats in marine environments.

SALIENT CHARACTERISTICS:

General The coating shall be readily dispersible by hand stirring to a uniform condition and shall be compatible with odorless mineral spirits. The coating shall brush and spray easily when thinned not more than 15% by volume with mineral spirits thinner and shall be capable of being applied at a wet film thickness of 10 mils. The dried film shall be free of brush marks, sags, runs, pinholes and other defects when applied to a dry thickness of 1.5 mm (0.006 in). The film shall dry smooth and have a uniform appearance.

Color The color shall be an approximate match to FED-STD-595 color number 27880 or 26270 as specified (ASTM D 1729).

Adhesion The coating shall tightly adhere to aged alkyd coatings and aged rust preventive coatings when the surface has been prepared by water blast cleaning and/or detergent washing. The coating shall dry to a soft non-brittle film[1].

Water resistance The enamel shall show no blistering or wrinkling after 24 hours immersion in distilled water (ASTM D 1308). Twenty-four hours after removal, the test film shall be equal in hardness, adhesion, general appearance to the unexposed panel, and shall retain a minimum of 90 percent of the initial 60° gloss (ASTM D 523)[1].

Accelerated Weathering After weathering 500 hours using fluorescent UV-condensation accelerated weathering apparatus (ASTM G 53)[1] and an exposure cycle of 240 minutes light at 60° C followed by 240 minutes condensation without light at 50° C, the test film shall meet the following requirements:

Requirement	Limits	Test Method
Chalking	Slight	ASTM D 659
60° gloss, percent of original retained	25 min	ASTM D 523

Salt Spray (Fog) Testing After 1000 hours (ASTM B 1417) the sample shall show no rusting or undercutting when evaluated in accordance with ASTM D 610[2].

Nonvolatile Vehicle - The nonvolatile vehicle shall be a phenolic resin modified with an inorganic corrosion inhibitor.

Pigment - The pigment shall be titanium dioxide, Type IV, ASTM D 476, and suitable color pigments for gray color. THE Coating shall contain no aluminum materials.

Volatile[3]- The volatile shall not be photochemically reactive as defined in Rule 102 of SCAQMD[3].

QUANTITATIVE REQUIREMENTS	LIMITS MIN.	LIMITS MAX.	ASTM TEST METHOD
60° gloss (5 Mil Dry Film - 72 hour dry)	---	10	D 523
Consistency, K.U.	100	120	D 562
Fineness of grind (Hegman)	2	---	D 1210
Dry time, set-to-touch, hours (5 mil dry film)	---	2	D 1640
Dry time, dry-hard, hours (5 mil dry film)	---	24	D 1640
Sag resistance	10	---	D 4400
Density, Weight per gallon, lbs	10.0	---	D 1475
Volatile Content, percent by weight	22	30	D 2369
Nonvolatile, percent by volume	59	---	D 2697
Flash point, °F	100	---	D 3278
Contrast ratio (at 5 mil dry film thickness)	1.00	---	D 2805

FSC 8010

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The issues of the ASTM test methods in effect on the date of the solicitation shall be used to determine compliance with these requirements.8010

Certification The contractor shall certify that the coating offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product sold in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under provisions of the contract.

Preservation, Packaging, Packing, Labeling and Marking The enamel shall be packaged in accordance with normal commercial practice and packed to assure acceptance by common carrier and provide protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with National Motor Freight Classification and Uniform Freight Classification. Shipping containers shall be marked as specified in the contract or order. Size shall be 5 gallon steel pail.

Notes ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

1. Test samples shall be prepared by brushing or spraying the coating to a dry film thickness of 5 mils over aged TT-E-490 silicone alkyd (1.5 mil dry film on steel panels weathered 168 hours as specified above).

2. Test samples shall be prepared by brushing or spraying the coating to a dry film thickness of 5 mils on rusted steel (Rust Gr D, SSPC VIS-1 Standard cleaned with hand tools in accordance with SSPC-SP-2)[4] and air dried 72 hours prior to test.

3. Information may be obtained from the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.

4. Information may be obtained from the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.

Preparing Activity
Navy -NAVSEA 070R

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